Lecture 20: Chebychev Nodes; Spline Spaces and B-Splines

Math 6610

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Today's Goals

- Finish Lecture 19: interpolation error (with proofs); error of splines
- Splines $\mathcal{S}_{k,\Delta}$ as a vector space of dimension $k+n-1$
- Define B-splines N_{ik} by the Cox–de Boor recursion

Interpolation Error: Statements

Theorem (*Interpolation Error I*). Let $f^{(n+1)}$ be continuous on $[a,b]$, and let $p_n \in \prod_n$ interpolate f at distinct nodes $x_0, \ldots, x_n \in [a,b]$. Then for every $x \in [a,b]$ there is $c = c(x) \in (a,b)$ with

$$e(x) := f(x) - p_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} \prod_{i=0}^{n} (x - x_i).$$

Lemma (*Bounding Lemma*). Let $n \geqslant 1$, $h = (b-a)/n$ and $x_i = a + ih$, $i = 0, \ldots, n$. Then for every $x \in [a,b]$

$$\prod_{i=0}^{n} |x - x_i| \leqslant \frac{h^{n+1}}{4}\, n!.$$

Remark. The theorem holds for any distinct nodes; the lemma is for equally spaced nodes only. Proofs on the next three slides.

Interpolation Error I: Proof

Proof. Idea: build a function with $n+2$ zeros, then use Rolle $n+1$ times. Write $\omega(t) := \prod_{i=0}^{n}(t - x_i)$.

1. Setup. If x is a node, both sides are 0 (any c works). Otherwise $\omega(x) \neq 0$; set

$$g(t) := e(t) - \frac{\omega(t)}{\omega(x)}\, e(x), \qquad t \in [a, b].$$

2. Zeros. $g = 0$ at $x_0, \ldots, x_n$ (as $e = \omega = 0$ there) and at $t = x$: $n+2$ distinct zeros in $[a, b]$.

3. Rolle. g has $n+1$ continuous derivatives. Between two zeros of a C^1 function lies a zero of its derivative. So g' has $n+1$ distinct zeros, …, $g^{(n+1)}$ has one, $c \in (a, b)$.

4. Differentiate $n+1$ times. $p_n^{(n+1)} = 0$ (degree $\leqslant n$) and $\omega^{(n+1)} = (n+1)!$ ($\omega = t^{n+1} + \ldots$):

$$0 = g^{(n+1)}(c) = f^{(n+1)}(c) - (n+1)!\,\frac{e(x)}{\omega(x)} \implies e(x) = \frac{f^{(n+1)}(c)}{(n+1)!}\,\omega(x).$$

■

Bounding Lemma: Proof

Proof of the Bounding Lemma. 1. Locate x. Choose $j \in \{0, \ldots, n-1\}$ with $x \in [x_j, x_{j+1}]$.
2. Nearest nodes. Two distances with sum h; their product is largest when both are $h/2$: $\left|(x - x_j)(x - x_{j+1})\right| \leqslant (h/2)^2$.
3. Other nodes. $|x - x_i| \leqslant (j + 1 - i)h$ for $i < j$, and $|x - x_i| \leqslant (i - j)h$ for $i > j + 1$. Multiply all $n + 1$ factors:

$$\prod_{i=0}^{n} |x - x_i| \leqslant \frac{h^2}{4} \cdot (j+1)!\, h^j \cdot (n-j)!\, h^{n-j-1} = \frac{h^{n+1}}{4}\, (j+1)!\, (n-j)!.$$

4. Factorials. $1 \leqslant j + 1 \leqslant n$, so $\binom{n+1}{j+1} \geqslant n + 1$ and

$$(j+1)!\, (n-j)! = \frac{(n+1)!}{\binom{n+1}{j+1}} \leqslant \frac{(n+1)!}{n+1} = n!.$$

■

Interpolation Error II

Corollary (*Interpolation Error II*). With f, p_n as in the theorem, nodes as in the lemma, and $M := \max_{[a,b]} \left|f^{(n+1)}\right|$:

$$\max_{[a,b]} |e(x)| \leqslant \frac{Mh^{n+1}}{4(n+1)}.$$

Proof of Interpolation Error II. M exists: $f^{(n+1)}$ is continuous on the closed interval. Insert $\left|f^{(n+1)}(c)\right| \leqslant M$ and the lemma into Interpolation Error I: $|e(x)| \leqslant \frac{M}{(n+1)!} \cdot \frac{h^{n+1}}{4} n! = \frac{Mh^{n+1}}{4(n+1)}$. ■

Remark. $M = \max_{[a,b]} \left|f^{(n+1)}\right|$ depends on n. If it grows fast with n, the bound need not tend to 0 as $n \to \infty$: Runge's example below.

Reading the Error Formula

$$e(x) = \underbrace{\frac{f^{(n+1)}(c)}{(n+1)!}}_{\text{how smooth } f \text{ is}} \underbrace{\omega(x)}_{\text{where the nodes are}}, \qquad \omega(x) = \prod_{i=0}^{n}(x - x_i).$$

- Bound for any nodes: with $M = \max_{[a,b]} \left|f^{(n+1)}\right|$,

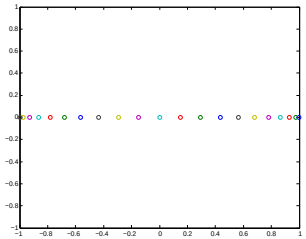
$$\max_{[a,b]} |e| \leqslant \frac{M}{(n+1)!} \max_{x \in [a,b]} |\omega(x)| .$$

- M is fixed by f. Only ω is ours to choose, through the nodes.
- Equally spaced, $n = 10$ on $[-1, 1]$: $\max |\omega| \approx 8.5 \cdot 10^{-3}$, reached in the first and last intervals (picture: “The Node Product”).
- Question: which nodes in $[-1, 1]$ make $\max_{[-1,1]} |\omega|$ as small as possible? Two remedies for Runge’s example: move the nodes (next), or piecewise polynomials on a fixed grid (splines).

Chebychev Nodes

Definition (*Chebychev nodes*). The Chebychev nodes of degree n in $[-1, 1]$ are

$$x_i = \cos\left(\pi \frac{2i+1}{2n+2}\right), \qquad i = 0, \ldots, n; \qquad \text{on } [a, b]: \ \tfrac{a+b}{2} + \tfrac{b-a}{2} x_i.$$



- Equally spaced angles on the half circle, projected down to the x-axis
- Not uniform (bunched at the ends) and not nested (new n, new nodes)

Chebychev Polynomials

Definition (*Chebychev polynomial*). For $m = 0, 1, 2, \ldots$ and $x \in [-1, 1]$, $T_m(x) := \cos(m \arccos x)$. Equivalently, with $x = \cos\theta$, $\theta \in [0, \pi]$: $T_m(\cos\theta) = \cos(m\theta)$.

Lemma. $T_0(x) = 1$, $T_1(x) = x$, and $T_{m+1}(x) = 2x\,T_m(x) - T_{m-1}(x)$ for $m \geqslant 1$. Hence, for $m \geqslant 1$, T_m is a polynomial of degree m with leading coefficient 2^{m-1}.

Proof. 1. Recurrence. The identity $\cos((m+1)\theta) + \cos((m-1)\theta) = 2\cos\theta\,\cos(m\theta)$ with $x = \cos\theta$ reads $T_{m+1} + T_{m-1} = 2x\,T_m$.
2. Degree, induction on m. Base: $T_1 = x$ has degree 1 and leading coefficient $1 = 2^0$. Step: if T_m has degree m, leading coefficient 2^{m-1}, and T_{m-1} has degree $m-1$, then $2x\,T_m - T_{m-1}$ has degree $m+1$ and leading coefficient $2 \cdot 2^{m-1} = 2^m$. ■

Example. $T_2 = 2x^2 - 1$, $T_3 = 4x^3 - 3x$. (Check: $2x(2x^2 - 1) - x = 4x^3 - 3x$.)

Zeros and Extremes of T_{n+1}

Proposition. Let $n \geqslant 0$.

(i) The zeros of T_{n+1} are exactly the $n+1$ Chebychev nodes $x_i = \cos\left(\pi \frac{2i+1}{2n+2}\right)$, $i = 0, \ldots, n$.

(ii) $|T_{n+1}(x)| \leqslant 1$ for $x \in [-1, 1]$, and at the $n+2$ points $y_j := \cos \frac{j\pi}{n+1}$, $j = 0, \ldots, n+1$, $T_{n+1}(y_j) = (-1)^j$.

Proof. Write $x = \cos\theta$; this is one-to-one from $\theta \in [0, \pi]$ onto $x \in [-1, 1]$.
(i) $\cos((n+1)\theta) = 0 \Leftrightarrow (n+1)\theta = (2i+1)\frac{\pi}{2}$; $\theta \in [0, \pi]$ allows exactly $i = 0, \ldots, n$. These $n+1$ zeros are distinct, and a polynomial of degree $n+1$ has no more.
(ii) $|\cos| \leqslant 1$, and $\theta_j = \frac{j\pi}{n+1}$ gives $\cos((n+1)\theta_j) = \cos(j\pi) = (-1)^j$. ∎

Remark. $1 = y_0 > y_1 > \cdots > y_{n+1} = -1$: T_{n+1} swings between $+1$ and -1 exactly $n+2$ times. This alternation is the key to the next proof.

Chebychev Nodes Are Optimal

Theorem. Let $x_0, \ldots, x_n$ be the Chebychev nodes and $\omega(x) = \prod_{i=0}^{n}(x - x_i)$.

(a) $\omega = 2^{-n}T_{n+1}$; hence $\max_{[-1,1]} |\omega| = 2^{-n}$.

(b) For any $z_0, \ldots, z_n \in \mathbb{R}$, $\max_{x\in[-1,1]} \prod_{i=0}^{n} |x - z_i| \geqslant 2^{-n}$.

Proof. 1. Part (a). ω and $2^{-n}T_{n+1}$ both have degree $n+1$, leading coefficient 1 ($2^{-n} \cdot 2^n$), and the same $n+1$ zeros (Proposition (i)). Their difference has degree $\leqslant n$ and $n+1$ zeros, so it is 0. By (ii), $\max |T_{n+1}| = 1$ (reached at $y_0 = 1$).

2. Part (b): assume the opposite. Let $q(x) := \prod_i (x - z_i)$ with $|q(x)| < 2^{-n}$ for all $x \in [-1, 1]$, and set

$$r := 2^{-n}T_{n+1} - q.$$

3. Degree. Both terms are $x^{n+1} +$ (lower powers); the x^{n+1} cancels: $r \in \prod_n$.

(continued on the next slide)

Chebychev Nodes Are Optimal (cont'd)

Proof (continued). 4. Signs. At the points y_j of (ii), $j = 0, \ldots, n+1$,

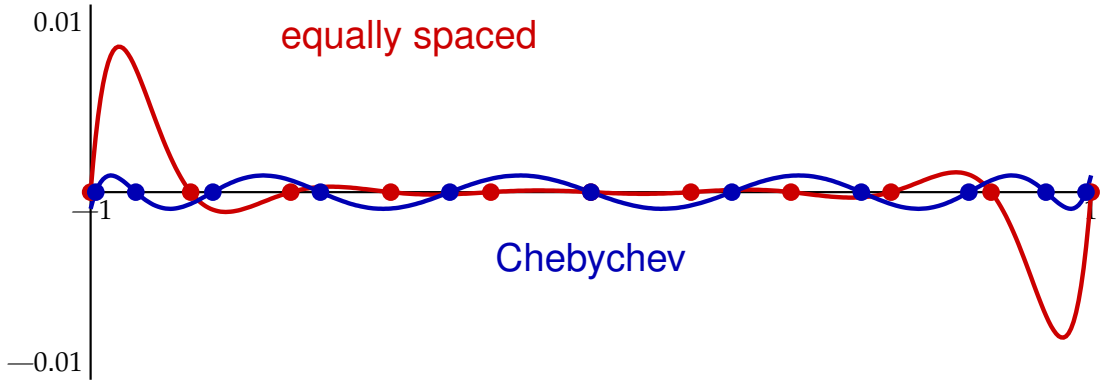
$$r(y_j) = (-1)^j 2^{-n} - q(y_j), \qquad \left|q(y_j)\right| < 2^{-n},$$

so $r(y_j)$ has the sign of $(-1)^j$: r alternates in sign at $n+2$ points.
5. Zeros. r is continuous; by the intermediate value theorem it has a zero in each of the $n+1$ disjoint intervals (y_{j+1}, y_j), $j = 0, \ldots, n$.
6. Contradiction. $r \in \prod_n$ with $n+1$ distinct zeros, so $r \equiv 0$. But $r(y_0) > 0$. ■

Remark. In words: no choice of $n+1$ nodes makes the factor $\max|\omega|$ in the error bound smaller than the Chebychev nodes do. The proof used only the alternation of T_{n+1} at $n+2$ points.

The Node Product: A Picture

Example. $\omega(x) = \prod_i (x - x_i)$ for $n = 10$:



- Equally spaced: $\max|\omega| \approx 8.5 \cdot 10^{-3}$, large near the ends.
- Chebychev: $\max|\omega| = 2^{-10} \approx 9.8 \cdot 10^{-4}$; $|\omega| = 2^{-10}$ at all 12 points y_j (equal ripples).

Error Bound with Chebychev Nodes

Corollary. Let $f^{(n+1)}$ be continuous on $[a,b]$, $M := \max_{[a,b]} \left|f^{(n+1)}\right|$, and let p_n interpolate f at the Chebychev nodes on $[a,b]$, $\hat{x}_i = \frac{a+b}{2} + \frac{b-a}{2} x_i$. Then

$$\max_{[a,b]} |f - p_n| \leqslant \frac{M}{(n+1)!} \cdot 2\left(\frac{b-a}{4}\right)^{n+1}.$$

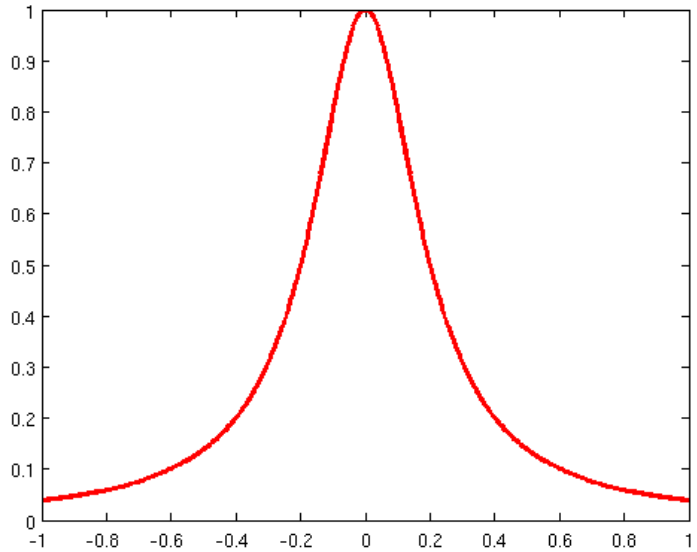
Proof. 1. Change of variable. Each $x \in [a,b]$ is $x = \frac{a+b}{2} + \frac{b-a}{2} t$ with $t \in [-1,1]$; then $x - \hat{x}_i = \frac{b-a}{2}(t - x_i)$.
2. Product. By the theorem, part (a),

$$\prod_{i=0}^{n} |x - \hat{x}_i| = \left(\frac{b-a}{2}\right)^{n+1} \prod_{i=0}^{n} |t - x_i| \leqslant \left(\frac{b-a}{2}\right)^{n+1} 2^{-n} = 2\left(\frac{b-a}{4}\right)^{n+1}.$$

3. Insert into the bound of "Reading the Error Formula". ■

Equally Spaced vs. Chebychev Nodes

High degree polynomials using equispaced points suffer from many oscillations



- Chebychev nodes: the error is distributed more evenly

Runge's Example: The Experiment

Interpolate $f(x) = 1/(1+25x^2)$ with degree n; measure the maximum error on a fine grid:

```
f  = @(x) 1./(1 + 25*x.^2);
xx = linspace(-1, 1, 20001);                    % fine grid
for n = [5 10 20]
  xe = linspace(-1, 1, n+1);                    % equally spaced
  xc = cos(pi*(2*(0:n) + 1)/(2*n + 2));         % Chebychev
  pe = polyfit(xe, f(xe), n);  pc = polyfit(xc, f(xc), n);
  ee = max(abs(f(xx) - polyval(pe, xx)));
  ec = max(abs(f(xx) - polyval(pc, xx)));
  fprintf('%3d  %9.3g  %9.3g\n', n, ee, ec)
end
```

- `polyfit` uses the monomial basis: for $n = 20$ (equally spaced) its matrix has $\kappa_2 \approx 8.3 \cdot 10^8$ (Lecture 17). The errors on the next slide are still correct to the digits shown.
- The maximum over a grid is a lower bound for the true maximum; 20001 points are enough here.

Runge's Example Explained

Example. $f(x) = 1/(1+25x^2)$ on $[-1,1]$; maximum error (computed):

n	5	10	20
equally spaced	0.43	1.92	59.8
Chebychev	0.56	0.11	0.015

- M is large: $f = \sum_{m\geqslant 0}(-25x^2)^m$ for $|x| < 1/5$, so $\left|f^{(m)}(0)\right|/m! = 5^m$ for even m; hence $M/(n+1)! \geqslant 5^{n+1}$ for odd n.
- Equally spaced: $|\omega|$ is large near the ends; the error there grows with n.
- Chebychev: the bound of the corollary is $\geqslant 5^{n+1}2^{-n}$ for odd n and grows, yet the observed error decreases. A bound is only an upper bound. (Fact, proof omitted: for every Lipschitz continuous f, the Chebychev interpolants converge to f uniformly on $[-1,1]$.)
- Rule: high degree $\Rightarrow$ Chebychev nodes; fixed grid $\Rightarrow$ splines.

Error of a Degree 1 Spline

Corollary. Let f'' be continuous on $[a, b]$, S the degree 1 spline with $S(t_i) = f(t_i)$, $i = 0, \ldots, n$, and $h := \max_i h_i$. Then

$$\max_{x\in[a,b]} |f(x) - S(x)| \leqslant \frac{h^2}{8} \max_{[a,b]} |f''| \qquad (O(h^2)\text{: halving } h \text{ divides the bound by } 4).$$

Proof. 1. One interval. For $x \in [t_i, t_{i+1}]$, S is the line through f at t_i, t_{i+1}. Interpolation Error I with $n = 1$ on $[t_i, t_{i+1}]$: for some $c \in (t_i, t_{i+1})$,

$$f(x) - S(x) = \frac{f''(c)}{2}(x - t_i)(x - t_{i+1}).$$

2. The product is largest at the midpoint: $|(x - t_i)(x - t_{i+1})| \leqslant (h_i/2)^2 \leqslant h^2/4$.

3. Combine: $|f(x) - S(x)| \leqslant \frac{1}{2} \cdot \frac{h^2}{4} \max_{[a,b]} |f''|$, for every x in every interval. ■

Error of a Cubic Spline

Theorem (*Hall–Meyer 1976; proof omitted*). Let $f^{(4)}$ be continuous on $[a,b]$, and S the fixed-slope cubic spline with $S(t_i) = f(t_i)$ and end slopes $f'(t_0)$, $f'(t_n)$. With $h := \max_i h_i$,

$$\max_{[a,b]} |f - S| \leqslant \frac{5}{384} h^4 \max_{[a,b]} \left| f^{(4)} \right| .$$

Example. $f = \sin$ on $[0, \pi/2]$, n equal intervals; maximum error (computed):

n	4	8	16	32	ratio
degree 1	$1.9 \cdot 10^{-2}$	$4.8 \cdot 10^{-3}$	$1.2 \cdot 10^{-3}$	$3.0 \cdot 10^{-4}$	≈ 4
natural	$7.7 \cdot 10^{-3}$	$1.9 \cdot 10^{-3}$	$4.7 \cdot 10^{-4}$	$1.2 \cdot 10^{-4}$	≈ 4
not-a-knot	$6.0 \cdot 10^{-4}$	$4.1 \cdot 10^{-5}$	$2.6 \cdot 10^{-6}$	$1.6 \cdot 10^{-7}$	≈ 16
fixed-slope	$6.3 \cdot 10^{-5}$	$3.9 \cdot 10^{-6}$	$2.4 \cdot 10^{-7}$	$1.5 \cdot 10^{-8}$	≈ 16

(Check, $n = 4$: $h^2/8 = 0.019$ and $\frac{5}{384} h^4 = 3.1 \cdot 10^{-4}$.)

Natural: only $O(h^2)$ here: it forces $S''(\pi/2) = 0$, but $\sin''(\pi/2) = -1$.

Measuring the Order in Matlab

If $E(h) \approx Ch^p$, halving h gives $E(h)/E(h/2) \approx 2^p$, so $p \approx \log_2\big(E(h)/E(h/2)\big)$.

```
a = 0;  b = pi/2;  xx = linspace(a, b, 20001);
ns = [4 8 16 32];  err = zeros(size(ns));
for m = 1:numel(ns)
  t = linspace(a, b, ns(m) + 1);
  S = spline(t, [cos(a) sin(t) cos(b)]);  % end slopes f'=cos
  err(m) = max(abs(sin(xx) - ppval(S, xx)));
end
p = log2(err(1:end-1)./err(2:end))          % observed order
```

Observed p for the table on the previous slide:

	degree 1	natural	not-a-knot	fixed-slope
p	2.0	2.0	3.9–4.0	4.0

(Not-a-knot: `spline(t, sin(t))`; natural: `natspline`.) Not-a-knot needs no end slopes and still gives order 4: this is why it is the default in `spline(t, y)`.

From Degree to Order

Given: $\Delta = \{x_0, x_1, \ldots, x_n\}$ with $a = x_0 < x_1 < \cdots < x_n = b$, $n \geqslant 1$. (These are the knots t_i of Lectures 17–19.)

Definition (*Spline of order k*). Let $k \geqslant 2$. S is a *spline of order k* on Δ if $S \in C^{k-2}[a, b]$ ($S, S', \ldots, S^{(k-2)}$ continuous) and, for $j = 1, \ldots, n$, the restriction of S to $[x_{j-1}, x_j]$ is in $\prod_{k-1}$ (polynomials of degree $\leqslant k-1$). $\mathcal{S}_{k,\Delta}$ denotes the set of all such splines.

Order k = degree $k-1$, the name used in Lecture 17 (each piece has k coefficients):

order k	degree $k-1$	name (Lectures 17–19)
2	1	degree 1 (linear) spline, C^0
3	2	degree 2 (quadratic) spline, C^1
4	3	cubic spline, C^2

Remark. $\mathcal{S}_{k,\Delta}$ is a real vector space: sums and real multiples of splines of order k are splines of order k. Also $\prod_{k-1} \subset \mathcal{S}_{k,\Delta}$.

The Dimension of $\mathcal{S}_{k,\Delta}$

Proposition. $\dim \mathcal{S}_{k,\Delta} = k + n - 1$.

Proof: the theorem "The Truncated Power Basis" below gives a basis with $k + n - 1$ elements. First, why this number:

Remark (*Count, as in Lecture 17*).

- On $[x_0, x_1]$: a polynomial of degree $\leqslant k - 1$, so k free coefficients.
- On each next interval: k coefficients, minus $k - 1$ matching conditions ($S, S', \ldots, S^{(k-2)}$ continuous at the knot): 1 new free coefficient.
- Total: $k + (n - 1)$.

The count is not a proof: it assumes the matching conditions are independent.

Remark (*Link to Lecture 18*). Interpolation gives $n + 1$ conditions, so $k + n - 1 - (n + 1) = k - 2$ conditions remain free. Cubic ($k = 4$): the 2 end conditions.

Truncated Powers

Definition (*Truncated power*). For a knot x_j and $k \geqslant 2$,

$$(x - x_j)_+^{k-1} := \big(\max\{x - x_j, 0\}\big)^{k-1} = \begin{cases} (x - x_j)^{k-1}, & x \geqslant x_j, \\ 0, & x < x_j. \end{cases}$$

Lemma. $(x - x_j)_+^{k-1} \in C^{k-2}(\mathbb{R})$; its $(k-1)$-st derivative jumps by $(k-1)!$ at x_j. So $(x - x_j)_+^{k-1} \in \mathcal{S}_{k,\Delta}$, $j = 1, \ldots, n-1$.

Proof. 1. Away from x_j, for $l = 0, \ldots, k-2$:

$$\frac{d^l}{dx^l}(x - x_j)_+^{k-1} = (k-1)(k-2)\cdots(k-l)\,(x - x_j)_+^{k-1-l}.$$

2. At x_j, induction on l: the difference quotient of $c\,(x - x_j)_+^{k-l}$ at x_j is $c\,\eta_+^{k-l}/\eta \to 0$ ($k - l \geqslant 2$). So the formula also holds at x_j, and is continuous.
3. Order $k-1$*:* 0 for $x < x_j$, $(k-1)!$ for $x > x_j$. ■

The Truncated Power Basis

Theorem. The $k+n-1$ functions

$$B = \{1, x, \ldots, x^{k-1}, (x-x_1)_+^{k-1}, \ldots, (x-x_{n-1})_+^{k-1}\}$$

form a basis of $\mathcal{S}_{k,\Delta}$ (each $S \in \mathcal{S}_{k,\Delta}$ is a combination of them in exactly one way).

Proof. $B \subset \mathcal{S}_{k,\Delta}$ by the lemma. *1. Independence: setup.* Suppose, for all $x \in [a,b]$,

$$S(x) := \sum_{l=0}^{k-1} \alpha_l x^l + \sum_{i=1}^{n-1} \beta_i (x-x_i)_+^{k-1} = 0.$$

2. Jump at x_j, $1 \leqslant j \leqslant n-1$. Near x_j the monomials and $(x-x_i)_+^{k-1}$, $i \neq j$, are polynomials, so their $(k-1)$-st derivatives do not jump there. By the lemma,

$$0 = S^{(k-1)}(x_j^+) - S^{(k-1)}(x_j^-) = \beta_j (k-1)!, \qquad \text{so } \beta_j = 0.$$

(continued on the next slide)

The Truncated Power Basis (cont'd)

Proof (continued). 3. Monomials. Then $\sum_l \alpha_l x^l = 0$ on $[a, b]$: a polynomial with infinitely many zeros, so all $\alpha_l = 0$. Hence B is linearly independent.
4. Spanning. Let $S \in \mathcal{S}_{k,\Delta}$. Let $p \in \prod_{k-1}$ be the polynomial with $p = S$ on $[x_0, x_1]$, and c_j the jump of $S^{(k-1)}$ at x_j ($S^{(k-1)}$ is constant on each interval). Set

$$R := S - p - \sum_{j=1}^{n-1} \frac{c_j}{(k-1)!}(x - x_j)_+^{k-1}.$$

5. $R = 0$, induction over the intervals. $R \in \mathcal{S}_{k,\Delta}$, $R = 0$ on $[x_0, x_1]$, and $R^{(k-1)}$ has no jumps left (the lemma removes c_j). If $R = 0$ on $[x_{j-1}, x_j]$, then $R, R', \ldots, R^{(k-1)}$ are 0 at x_j from the left, hence from the right; R is in $\prod_{k-1}$ on $[x_j, x_{j+1}]$, so by Taylor's formula $R = 0$ there. So $S = p + \sum_j \frac{c_j}{(k-1)!}(x - x_j)_+^{k-1} \in \operatorname{span} B$. ■

Truncated Powers in Practice

Interpolate with the not-a-knot cubic (Lecture 18), $n \geqslant 3$: its knots are $x_2, \ldots, x_{n-2}$, so the basis is $1, x, x^2, x^3, (x - x_j)_+^3$, $j = 2, \ldots, n-2$: $n+1$ functions, $n+1$ data.

```
n = 20;  x = linspace(0, 1, n+1)';              % equally spaced
T = [x.^0  x  x.^2  x.^3  max(x - x(3:n-1)', 0).^3];
cond(T, inf)
```

n	5	10	20	40	80
$\kappa_\infty(T)$	$2.9 \cdot 10^3$	$4.2 \cdot 10^4$	$4.9 \cdot 10^5$	$6.4 \cdot 10^6$	$9.2 \cdot 10^7$

- **Ill-conditioned:** for nearby knots, $(x - x_j)_+^3$ and $(x - x_{j+1})_+^3$ nearly coincide; the columns of T are nearly parallel. About 8 digits can be lost at $n = 80$ (Lecture 15: relative error up to $\approx \kappa \cdot \epsilon_{\text{mach}}$).
- **Global support:** $(x - x_j)_+^3 \neq 0$ on all of $(x_j, b]$: T is dense below the diagonal, and each evaluation of $S(x)$ touches $O(n)$ terms.

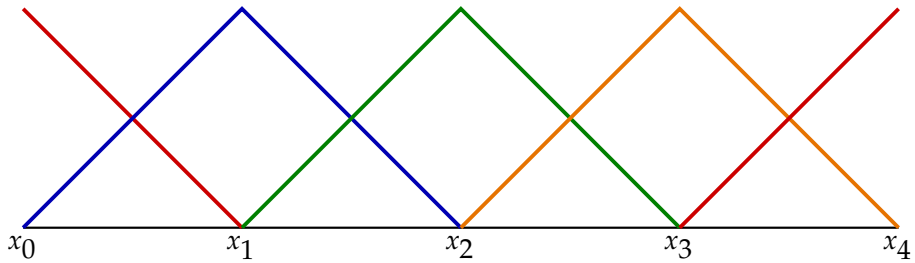
What We Want Instead

Wish list for a basis of $\mathcal{S}_{k,\Delta}$:

- *local:* each basis function is nonzero on at most k consecutive intervals;
- *nonnegative*, and *summing to* 1 at every $x \in [a, b]$;
- *well conditioned*, with cheap and stable evaluation.

The *B-splines* have the first two (proved today) and the third (measured in Lecture 21).

Example ($k = 2$: *the hat functions*). On $x_0, \ldots, x_4$: N_{i2} is 1 at x_{i-1}, 0 at the other knots, linear in between (precise definition: two slides on):



$$S(x) = \sum_{i=1}^{n+1} y_{i-1} N_{i2}(x) \quad \text{satisfies} \quad S(x_{i-1}) = y_{i-1},\ i = 1, \ldots, n+1.$$

The coefficients are the data: no linear system at all.

Knots for B-Splines

B-splines are built on a *knot sequence*, in which a value may repeat:

$$\tau_1 \leqslant \tau_2 \leqslant \cdots \leqslant \tau_l, \qquad \tau_1, \tau_2, \ldots, \tau_l \in \mathbb{R}. \tag{1}$$

Definition (*Multiplicity*). The *multiplicity* of a knot value is the number of times it appears in (1).

For $\mathcal{S}_{k,\Delta}$: interior knots once, end points k times ("clamped"):

$$\overbrace{\underbrace{x_0}_{\tau_1=\cdots=\tau_k}}^{k \text{ times}} < \underbrace{x_1}_{\tau_{k+1}} < \cdots < \underbrace{x_{n-1}}_{\tau_{k+n-1}} < \overbrace{\underbrace{x_n}_{\tau_{k+n}=\cdots=\tau_{2k+n-1}}}^{k \text{ times}}$$

$$l = 2k + n - 1, \qquad l - k = k + n - 1 = \dim \mathcal{S}_{k,\Delta}.$$

Example. $k = 2$, $n = 4$: $\tau = (x_0, x_0, x_1, x_2, x_3, x_4, x_4)$, $l = 7$, $l - k = 5$ hat functions. The doubled end points produce the two half hats.

B-Splines: Definition

Definition (*B-splines; Cox–de Boor recursion*). The B-splines N_{ij} (index i, order j) of the knots (1) are defined by recursion on j (each order blends two neighbours of the order below, with weights linear in x):

$$N_{i1}(x) := \begin{cases} 1, & \tau_i \leqslant x < \tau_{i+1}, \\ 0, & \text{otherwise}, \end{cases} \qquad i = 1, 2, \ldots, l-1,$$

and for $j = 2, 3, \ldots, k$:

$$N_{ij}(x) := \frac{x - \tau_i}{\tau_{i+j-1} - \tau_i} N_{i,j-1}(x) + \frac{\tau_{i+j} - x}{\tau_{i+j} - \tau_{i+1}} N_{i+1,j-1}(x), \qquad i = 1, 2, \ldots, l-j. \quad (2)$$

Conventions:

1. A term whose denominator is 0 is set to 0 ("$0/0 = 0$"). Nothing is lost: e.g. $\tau_i = \tau_{i+j-1}$ makes $N_{i,j-1} \equiv 0$ (property (a) below).
2. $N_{ik}(\tau_l) := \lim_{x \to \tau_l^-} N_{ik}(x)$, so that the right end point $b = \tau_l$ is covered.

Orders 1 and 2 by Hand

Take equally spaced knots $\tau_i = i$ (a long enough finite piece; no repeated knots). Order 1: boxes,

$$N_{i1}(x) = 1 \text{ for } i \leqslant x < i+1, \quad 0 \text{ otherwise.}$$

Order 2: (2) with $j = 2$, both denominators $= 1$:

$$N_{i2}(x) = (x-i)\,N_{i1}(x) + (i+2-x)\,N_{i+1,1}(x) = \begin{cases} x-i, & i \leqslant x < i+1, \\ i+2-x, & i+1 \leqslant x < i+2, \\ 0, & \text{otherwise.} \end{cases}$$

The hat function: up-ramp times the left box, plus down-ramp times the right box.

Remark. The weight $\frac{x-\tau_i}{\tau_{i+j-1}-\tau_i}$ rises linearly from 0 to 1 across $[\tau_i, \tau_{i+j-1}]$, where $N_{i,j-1}$ can be nonzero; the other weight falls from 1 to 0 across $[\tau_{i+1}, \tau_{i+j}]$. Each order widens the support (the set where the function is nonzero) by one interval.

Order 3 by Hand

Same knots, $j = 3$, both denominators $= 2$. With $s = x - i$:

$$N_{i3}(x) = \tfrac{s}{2} N_{i2}(x) + \tfrac{3-s}{2} N_{i+1,2}(x) = \begin{cases} \frac{1}{2}s^2, & 0 \leqslant s < 1, \\ \frac{1}{2}(-2s^2 + 6s - 3), & 1 \leqslant s < 2, \\ \frac{1}{2}(3-s)^2, & 2 \leqslant s < 3, \\ 0, & \text{otherwise.} \end{cases}$$

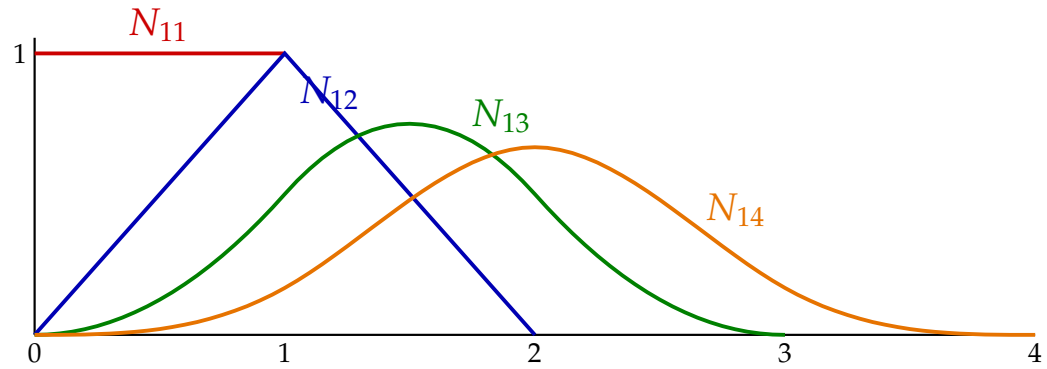
(Middle piece: $\frac{s}{2}(2-s) + \frac{3-s}{2}(s-1)$.)

- C^1 at $s = 1$: values $\frac{1}{2} = \frac{1}{2}$, slopes $1 = 1$; same at $s = 2$ by symmetry ($s \mapsto 3 - s$).
- At $x = 1.5$ only $N_{-1,3}, N_{0,3}, N_{1,3}$ are nonzero ($s = 2.5, 1.5, 0.5$):

$$\tfrac{1}{8} + \tfrac{3}{4} + \tfrac{1}{8} = 1.$$

Orders 1 to 4 on Equally Spaced Knots

$N_{11}, N_{12}, N_{13}, N_{14}$ for $\tau_i = i$: supports $[0,1]$, $[0,2]$, $[0,3]$, $[0,4]$.



- Each order: one more interval of support, one more continuous derivative, lower and wider.
- N_{14} is the cubic B-spline: C^2, and at the ends of its support $N_{14} = N'_{14} = N''_{14} = 0$.

B-Splines in Matlab: The Recursion

(2) translates line by line (`tau(i)` $= \tau_i$):

```
function v = bspl(tau, i, j, x)
% N_{ij}(x) by the recursion; x may be a vector
if j == 1
  v = double(tau(i) <= x & x < tau(i+1));  return
end
v  = zeros(size(x));
d1 = tau(i+j-1) - tau(i);  d2 = tau(i+j) - tau(i+1);
if d1 > 0                                % d1 = 0: term is 0
  v = v + (x - tau(i))/d1 .* bspl(tau, i, j-1, x);
end
if d2 > 0
  v = v + (tau(i+j) - x)/d2 .* bspl(tau, i+1, j-1, x);
end
```

B-Splines in Matlab: The Recursion (cont'd)

Plot the clamped cubic B-splines on $x_0, \ldots, x_5 = 0, 1, \ldots, 5$ (picture: Lecture 21):

```
tau = [0 0 0 0 1 2 3 4 5 5 5 5];  k = 4;    % 8 B-splines
xx = linspace(0, 5 - 1e-12, 501);            % x < b = 5
for i = 1:numel(tau)-k
  plot(xx, bspl(tau, i, k, xx)); hold on
end
```

Check the partition of unity numerically:

```
s = 0;  for i = 1:8, s = s + bspl(tau, i, k, xx); end
max(abs(s - 1))                               % about 1e-16
```

Remark. Correct, but wasteful: one call for N_{ik} makes up to $2^k - 1$ calls and recomputes the same lower-order values. Next lecture: share them.

Five Properties

Proposition. Let (1) hold and $1 \leqslant i \leqslant l-k$.

(a) *Support:* $N_{ik}(x) = 0$ for $x \notin [\tau_i, \tau_{i+k})$.

(b) *Nonnegative:* $N_{ik}(x) \geqslant 0$ for all $x \in \mathbb{R}$.

(c) *Partition of unity:* if $\tau_k < \tau_{l-k+1}$, then $\sum_{i=1}^{l-k} N_{ik}(x) = 1$ for all $x \in [\tau_k, \tau_{l-k+1})$.

(d) *Smoothness:* at a knot value of multiplicity $m \leqslant k$, N_{ik} is C^{k-1-m} (for $m = k$: it may jump); proof omitted.

(e) *Piecewise polynomial:* $N_{ik} \in \prod_{k-1}$ on each $[\tau_j, \tau_{j+1})$, $j = 1, \ldots, l-1$.

Remark (*Clamped knots*). For the knots of "Knots for B-Splines": $[\tau_k, \tau_{l-k+1}] = [a, b]$; (c) holds on all of $[a, b]$ (at b by convention 2); the interior knots are simple ($m = 1$), so by (d) and (e) every N_{ik} is in $\mathcal{S}_{k,\Delta}$.

Properties (a), (b), (e): Proof

Proof. Induction on the order $j = 1, \dots, k$, for all i at once: we show $N_{ij} = 0$ outside $[\tau_i, \tau_{i+j})$, $N_{ij} \geqslant 0$, and $N_{ij} \in \prod_{j-1}$ on each $[\tau_m, \tau_{m+1})$.
1. Base $j = 1$. N_{i1} is 1 on $[\tau_i, \tau_{i+1})$ and 0 elsewhere: all three hold.
2. Support. By the induction hypothesis $N_{i,j-1}$ vanishes outside $[\tau_i, \tau_{i+j-1})$ and $N_{i+1,j-1}$ outside $[\tau_{i+1}, \tau_{i+j})$; both lie in $[\tau_i, \tau_{i+j})$, so by (2) N_{ij} vanishes outside it.
3. Sign. Where $N_{i,j-1}(x) \neq 0$: $x \in [\tau_i, \tau_{i+j-1})$, so its weight is $\geqslant 0$. Where $N_{i+1,j-1}(x) \neq 0$: $x \in [\tau_{i+1}, \tau_{i+j})$, so its weight is > 0. With $N_{\cdot,j-1} \geqslant 0$ (induction hypothesis), $N_{ij} \geqslant 0$.
4. Degree. On $[\tau_m, \tau_{m+1})$, each term of (2) is (degree $\leqslant 1$) $\times$ (degree $\leqslant j-2$), so $N_{ij} \in \prod_{j-1}$ there. ∎

Remark. A cubic B-spline ($k = 4$) is nonzero on at most 4 intervals. This locality is what makes every later computation cheap.

Partition of Unity: Proof

Write $\omega_{ij}(x) := \dfrac{x - \tau_i}{\tau_{i+j-1} - \tau_i}$ ($:= 0$ if the denominator is 0). Then

$$\frac{\tau_{i+j} - x}{\tau_{i+j} - \tau_{i+1}} = 1 - \omega_{i+1,j}(x) \quad \text{if } \tau_{i+j} > \tau_{i+1},$$

so (2) reads

$$N_{ij} = \omega_{ij} N_{i,j-1} + (1 - \omega_{i+1,j}) N_{i+1,j-1}$$

(also if $\tau_{i+j} = \tau_{i+1}$: then $N_{i+1,j-1} \equiv 0$ by (a)).

Proof of (c). Fix $x \in [\tau_k, \tau_{l-k+1})$ and let m be the index with $\tau_m \leqslant x < \tau_{m+1}$; then $k \leqslant m \leqslant l - k$. Induction on $j = 1, \ldots, k$:

$$\sum_{i=1}^{l-j} N_{ij}(x) = 1.$$

1. Base $j = 1$. The boxes $[\tau_i, \tau_{i+1})$ are disjoint; only $N_{m1}(x) = 1$ is nonzero.
(continued on the next slide)

Partition of Unity: Proof (cont'd)

Proof (continued). 2. Sum the recursion over $i = 1, \ldots, l-j$; shift $i+1 \to i$ in the second part:

$$\sum_{i=1}^{l-j} N_{ij} = \sum_{i=1}^{l-j} \omega_{ij} N_{i,j-1} + \sum_{i=2}^{l-j+1} (1-\omega_{ij}) N_{i,j-1}.$$

3. Combine the terms with the same $N_{i,j-1}$, $2 \leqslant i \leqslant l-j$: $\omega_{ij} + (1-\omega_{ij}) = 1$. Hence

$$\sum_{i=1}^{l-j} N_{ij}(x) = \sum_{i=1}^{l-j+1} N_{i,j-1}(x) - (1-\omega_{1j}) N_{1,j-1}(x) - \omega_{l-j+1,j} N_{l-j+1,j-1}(x).$$

4. The two end terms vanish at x. By (a), $N_{1,j-1}(x) \neq 0$ needs $x < \tau_j \leqslant \tau_k \leqslant \tau_m$, and $N_{l-j+1,j-1}(x) \neq 0$ needs $x \geqslant \tau_{l-j+1} \geqslant \tau_{l-k+1} \geqslant \tau_{m+1}$. Neither holds.
5. Induction hypothesis. The remaining sum is 1 (order $j-1$). ■

Remark. With the end knots repeated k times, $[\tau_k, \tau_{l-k+1}] = [a, b]$: the B-splines sum to 1 on the whole interval.

What the Properties Buy

For $S(x) = \sum_i \gamma_i N_{ik}(x)$ on clamped knots:

- **(a) local support** $\Rightarrow$ at each x at most k terms are nonzero: evaluation in $O(k^2)$ operations, banded interpolation matrices, local edits (next lecture).
- **(b)** $N_{ik} \geqslant 0$ **and (c)** $\sum_i N_{ik} = 1 \Rightarrow S(x)$ is a weighted average (convex combination: weights $\geqslant 0$, sum 1) of the coefficients:

$$\min_i \gamma_i \leqslant S(x) \leqslant \max_i \gamma_i, \qquad x \in [a, b].$$

  No overshoot beyond the coefficients; $\max_{[a,b]} |S| \leqslant \max_i |\gamma_i|$.
- **(d) smoothness and (e) piecewise** $\prod_{k-1} \Rightarrow$ each $N_{ik} \in \mathcal{S}_{k,\Delta}$; next lecture: they form a *basis*.

B-splines (and their rational versions, NURBS) are the standard representation of curves in CAD and computer graphics (Lecture 17: fonts, the teapot).

Looking Ahead

- **Next lecture:** B-Splines: Evaluation and Interpolation